Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[a4paper]geometry mtsummit2023 times url latexsym natbib layout [hang,marginal]footmisc

Formatting Your Paper
for the MT Summit XIX (2023) Conference

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Ringkasan

This document contains the instructions for preparing a camera-ready manuscript for the proceedings of the 19th Machine Translation Summit (MT Summit XIX) in 2023. The document itself conforms to its own specifications and is therefore an example of what your manuscript should look like. Authors are asked to conform to all the directions reported in this document.

1 Credits

This text format is derived from that of the Journal of Evolutionary Computation.[1] The instructions in this document are derived from those of the MT Summit 2021, which was the latest in a long line of adaptations: MT Summit 2019, MT Summit 2017, MT Summit 2015, MT Summit 2013, EAMT 2011, 2010 and 2009, Coling 2008, ACL-07, Coling/ACL-06, EACL-06, ACL-05 and EACL-03. All these were based on the formats of earlier ACL and EACL Conference proceedings. Those versions were written by several people, including John Chen, Henry S. Thompson and Donald Walker.

2 Introduction

The following formatting instructions are directed to authors of papers accepted for publication in the proceedings of 19th Machine Translation Summit (MT Summit XIX) in 2023, including the main conference, workshops, posters and demos. See also the conference website for additional advice and information regarding submission. All authors are required to adhere to these specifications. Authors are required to submit their papers in PDF (Portable Document Format). Papers are designed to be printed on **A4 paper**. Authors from countries in which access to word processing systems is limited should contact the appropriate program chair(s) as soon as possible.

3 General Instructions

Manuscripts must be in single-column format. The title must be centered at the top of the first page. Authors' names, email addresses and complete mailing addresses must appear below the title, as in the current document. **Type single-spaced.** Start all pages directly under the top margin. See the guidelines later regarding formatting the first page.

Unless otherwise specified, the maximum length of a manuscript is 10 (ten) pages, plus unlimited pages for references (see Section 4 for additional information on the maximum number of pages).

3.1 Electronically-available resources

This description is provided in LaTeX2e (`mtsummit2023.tex`) along with the LaTeX2e style file used to format it (`mtsummit2023.sty`) and bibliography (BibTeX) example file (`mtsummit2023.bib`);

[1] Originally written by Darrell Whitley, and only later modified by Marc Schoenauer, especially regarding the bibliography style.

and in PDF format (`mtsummit2023.pdf`). These files are all available on the conference website (`https://mtsummit2023.scimeeting.cn/en/web/index/`) and Overleaf (`https://www.overleaf.com/latex/templates/mt-summit-2023-template/knrrcnxhkqxd`). There is also a Microsoft Word document template (`mtsummit2023.docx`) available at the same URL. We strongly recommend the use of these style files, which have been appropriately tailored for the proceedings.

3.2 Format of Electronic Manuscript

For the production of the electronic manuscript you must use Adobe's Portable Document Format (PDF). This format can be generated directly from LaTeXsource files, using program pdfLaTeX, or it can be generated from postscript files. On Unix systems, you can use `ps2pdf` for this purpose. Recent versions of Microsoft Windows can produce PDF directly (File>Save As>Save As Type: PDF). Otherwise, you can use Adobe's Distiller or GSview (File>Convert>pdfwrite); if you have *cygwin* installed, you can use *ps2pdf*. Note that some word processing programs generate PDF which may not include all the necessary fonts (esp. tree diagrams, symbols). When you print or create the PDF file, there is usually an option in your printer setup to include none, all or just non-standard fonts. Please make sure that you select the option of including ALL the fonts. *Before sending it, test your PDF by printing it from a computer different from the one where it was created.* Moreover, some word processors may generate very large postscript / PDF files, where each page is rendered as an image. Such images may reproduce poorly. In this case, try alternative ways to obtain the postscript and / or PDF. One way on some systems is to install a driver for a postscript printer, send your document to the printer specifying "Output to a file", then convert the file to PDF.

It is of utmost importance to specify the **A4 format** (21.0 cm x 29.7 cm) / (8.3 in x 11.7 in) when formatting the paper. When working with `dvips`, for instance, one should specify `-t a4`.

Print-outs of the PDF file on A4 paper should look like the present document, which conforms to the formatting requirements. *Note that in order for your paper to print correctly, you should disable centering and scale-to-fit options on your printer.* If you cannot meet the above requirements about the production of your camera-ready paper, please contact the program chairs as soon as possible.

3.3 Layout

Format manuscripts in a single column, in the manner these instructions are formatted. The exact dimensions for a page on A4 paper are:

- Left margin: 3.9 cm (1.54 in)
- Right margin: 3.8 cm (1.5 in)
- Top margin: 4.5 cm (1.77 in)
- Bottom margin: 4.2 cm (1.65 in)
- Text width: 13.3 cm (5.25 in)
- Text height: 21.0 cm (8.25 in)

Pages should not be numbered. In LaTeX, this can be achieved by inserting

```
\pagestyle{empty}
```

after the `\maketitle` command.

3.4 Fonts

For uniformity, Adobe's **Times Roman** font should be used. In LaTeX2e this is accomplished by putting

```
\usepackage{times}
\usepackage{latexsym}
```

in the preamble. If Times Roman is unavailable, use **Computer Modern Roman** (LaTeX2e's default). Note that the latter is about 10% less dense than Adobe's Times Roman font.

3.5 The First Page

Draw a horizontal line the full width of the text above the title, and another one below the authors' names and affiliations. Center the title across the top of the page. Authors' names appear under the title, one per line, with the author name in boldface, flush to the left and corresponding email flush to the right. Affiliations and mail addresses appear under each corresponding author name, flush to the left. Do not use footnotes for affiliations. Do not include the paper ID number assigned during the submission process. Use the single-column format throughout the text.

Title: Place the title centered at the top of the first page, in a 18-point bold font. Long titles should be typed on two lines without a blank line intervening. Approximately, put the title at 3.8 cm (1.5 in) from the top of the page, followed by a blank line, then the authors' names in 11-point bold font, and the affiliations in 11-point regular font on the following line. Do not use only initials for given names (middle initials are allowed). Avoid capitalizing last names. The affiliation should contain the author's complete address, and if possible an electronic mail address. Leave about 1 cm (0.4 in) between the affiliation and the body of the first page.

Abstract: Type the abstract at the beginning of the first column. The width of the abstract text should be smaller than the width of the columns for the text in the body of the paper by about 1 cm (0.4 in) on each side. The word **Abstract** should be in 9-point bold font above the body of the abstract, which should be in 9-point font. The abstract should be a concise summary of the general thesis and conclusions of the paper. It should be no longer than 200 words.

Text: Begin typing the main body of the text immediately after the abstract. Use 10 point font for text. **Indent** when starting a new paragraph, except for the first paragraph following each heading.

3.6 Sections

Headings: Type and label section and subsection headings in the style shown on the present document. Use numbered sections (Arabic numerals) in order to facilitate cross references. Number subsections with the section number and the subsection number separated by a dot, in Arabic numerals. Do not number subsubsections. Use 9-point bold font for subsection headings and 11-point bold font for section headings.

Citations: Citations within the text appear in parentheses as Smith or, if the author's name appears in the text itself, as [**?**]. Citations in parentheses should not be used as linguistic phrases; for example, instead of "Smith argues that ..." say "[**?**] argues that ...". Treat double authors as in DD15, but write as in PDC when more than two authors are involved. Append lowercase letters to the year in cases of ambiguity as in JonesFirst. Collapse multiple citations in

parenthesis as in Smith,JonesFirst and like this for multiple citations with the same-named author: Smith,SmithConc,JonesFirst,JonesSecond

References: Gather the full set of references together under the heading **References**; place the section before any Appendices, unless they contain references. Arrange the references alphabetically by the first author's last-name, rather than by order of occurrence in the text, and invert the first-name and last-name of the first author (only). Provide as complete a citation as possible, using a consistent format, such as the one for *North American Opera*. Use of full names for authors rather than initials is preferred. Use full names for journals and conferences, not abbreviations (for example "45th Meeting of the Association for Computational Linguistics", not "ACL07").

The LaTeX2e and `apalike` BibTeX style files provided roughly fit the American Psychological Association format, allowing regular citations, short citations and multiple citations as described above.

Appendices: Appendices, if any, directly follow the text and the references (but see above). Letter them in sequence and provide an informative title: **Appendix A. Title of Appendix**.

Acknowledgement section should go as a last section immediately *before the references.* Do not number the acknowledgement section.

3.7 Footnotes

Footnotes: Put footnotes at the bottom of the page and use 9-point font. They may be numbered or referred to by asterisks or other symbols.[2] Footnotes should be separated from the main text by a line.[3]

3.8 Copyright

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will be added to the bottom of all papers when proceedings are edited. If, for some reason, this copyright notice is not adequate for your paper, please contact the program chairs as soon as possible.

3.9 Graphics

Illustrations: Place figures, tables, and photographs in the paper near where they are first discussed, rather than at the end, if possible.

[2] This is how a footnote should appear.
[3] Note the line separating the footnotes from the text.

Captions: Provide a caption for every illustration; number each one sequentially in the form: "Figure 1. Caption of the Figure.", "Table 1. Caption of the Table." Type the captions of the figures and tables below the body, using 10-point text.

4 Length of Submission

Unless otherwise specified, the maximum length is 10 (ten) pages, plus unlimited pages for references. The page limit should be observed strictly. All illustrations and appendices must be accommodated within these page limits, following the formatting instructions given in the present document.